

Curriculum Vitae

Florian Froment

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EDUCATION	INSTITUT CATHOLIQUE D'ARTS ET MÉTIERS Master of science, Department of Mechanical Engineering	Lieusaint, FR	2024-2027
	WREXHAM UNIVERSITY Master of science, Engineering Research Semester, Electric Propulsion & Composite Structures	Wrexham, UK	2026-2027
	POLITECHNIKA ŚLĄSKA (5/5 GPA) Master of science, Aeronautical Engineering & Energy Transition	Gliwice, PL	2025-2026
	UNIVERSITY OF TOULOUSE - Jean Jaurès (Top 5% GPA) Bachelor of science, Mechanical Engineering & High Precision Machinery Production	Figeac, FR	2021-2024

PROFESSIONAL EXPERIENCE

FIVES MACHINING - Capdenac-Gare, France

Project Manager Engineer | Sept 2026 - Present

Mechanical & Hydraulic Design Engineer | Sept 2025 - Sept 2026

Mechanical & Hydraulic Design Technician | Sept 2023 - Sept 2025

- Designed and integrated mechanical assemblies for large-scale 5-axis milling machines (up to 40-ton structures)
- Engineered hydraulic systems operating up to 300 bar ensuring pressure stability
- Reduced integration time by 65% through component standardization
- Improved maintenance accessibility decreasing downtime by 30%
- Produced 300+ detailed manufacturing and assembly drawings
- Proposed new features for R&D and manufacturing of carbon parts

KM-RoBoTa - Lausanne, Switzerland

Robotics / R&D Engineer | June 2025 - Sept 2025

- Designed biomimetic quadruped robotic architecture reducing structural mass by 20% with an increased rigidity
- Created a new type of trunk-hind legs and joints for robots (23 joints) with an increased torque to weight ratio
- Engineered lightweight flexible arm joint systems increasing range of motion by 75%, replication of the human range
- Developed AI-supported validation rigs reducing testing cycles using ROS2

HYDRELEC-TECHNIC - France

Hydraulic Technician | July 2022 - Aug 2022

- Installed and calibrated hydraulic systems across nuclear, aerospace, defense, and hydropower sectors
- Reduced material waste by 12% through optimized component selection
- Designed custom tools reducing installation time by 40%

RTX - COLLINS AEROSPACE - Figeac, France

Methods Engineering Intern | Feb 2023 - Apr 2023

- Implemented REACH-compliant industrial washing process eliminating regulatory risk
- Reduced part preparation time by 18% with a reduced amount of failure of cleaning steps
- Generated 30+ aerospace certifications manufacturing and process documents
- Realized quality control procedures with decision making processes.

AIRBUS HELICOPTERS - Paris-Le Bourget, France

Industrial Technician | Nov 2021 - Dec 2021

- Commissioned composite blade molding equipment improving cycle reliability by 17%
- Optimized hydraulic group calibration improving flow stability by 10%
- Troubleshoot the key point of failure of a RTM machine

ARTICLES - Publications & Preprints

Research Master Thesis - Wrexham University (Incoming Spring 2026)

Hydrogen from Waste in Households - 05/2026

Design of a Local BioMass-Based Energy System - 06/2026 - XIV EPAE Conference on the 04/12/2026

CERTIFICATIONS

- TOEIC - 910/990
- CAPM® - PMI (Incoming 2027)
- Lean Manufacturing & Continuous Improvement
- Hydraulic Systems Installation & Safety Certification

SELECTED - Invited talk & forum

SIANE 2024, 20th Techninov, 55th SIAE, ACTEMIUM 5.0, World Bank Group Youth Summit,

Csat interview on the Dyson Award, XIV EPAE Conference

Hobbies: Karting, Entrepreneurship (idea thinking), Fed watcher, Sartorial Art

Tools: Microsoft 365 & Copilot, CATIA V5 (+4500 h), Fusion 360 (+1000 h), SAP & TeamCenter (+200 h)

Languages: French - C2 (Native), English - C1 (Professional use), Spanish - B1, Polish - A2

REFERENCES - Recommendation letters available upon request

Abd Baghdad Ph.D. MSc.

Researcher
Plateforme CARMA
Chercheur Associé au laboratoire
PIMM, Cnam, ENSAM et CNRS
(UMR 8006)

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Kamilo Melo Ph.D. MSc.

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Energy Engineering -
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